

COURSE: EEE 4484 (DIGITAL ELECTRONICS & PULSE TECHNIQUES LAB.)
EXPERIMENT NO. 04
STUDY THE RESPONSE OF A SAMPLE-AND-HOLD (S/H) CIRCUIT

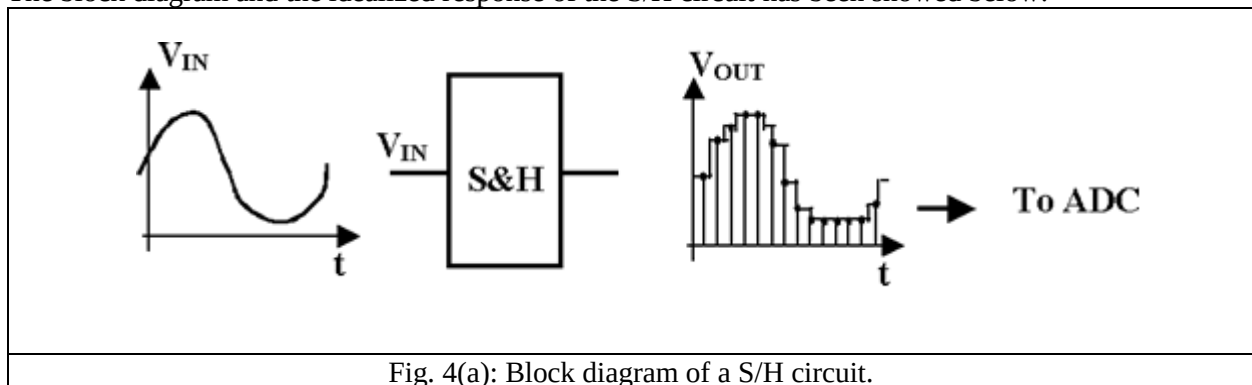
OBJECTIVE:

1. To study the response of a sample-and-hold (S/H) circuit.
2. Study the pin configuration of 555 Timer IC and Op-Amp (LM 741) IC and their uses in S/H circuit.

INTRODUCTION:

Sample-and-Hold (S/H) circuit is used to sample an analog input voltage for a very short period (several milliseconds to microseconds) and to hold on its last sampled value until the input is sampled again. The period may be from a few milliseconds to several seconds. The sample-and-hold (S/H) circuit has a third terminal for the logic S/H command in addition to the terminal for V_s and V_o . This command is issued to the circuit in the form of a pulse.

The block diagram and the idealized response of the S/H circuit has been showed below:



The circuit has two mode of operation depending on the logic level of the S/H command. Upon receiving the S/H pulse, circuit swings V_o toward V_s and then forces V_o to follow or track V_s for the remainder of the pulse. This is called the track mode. After the S/H pulse is removed, the circuit holds V_o at the value of V_s at the instant of pulse deactivation. This is called the hold mode.

A periodic array S/H pulse train allows the entire analog signal to be sampled and a discrete version of analog signal is reproduced at the output of the S/H circuit. For the through replication of the analog signal without distortion or interference known as aliasing, the sampling frequency, f_s of the S/H pulse train should be twice or more than the bandwidth of the input analog signal.

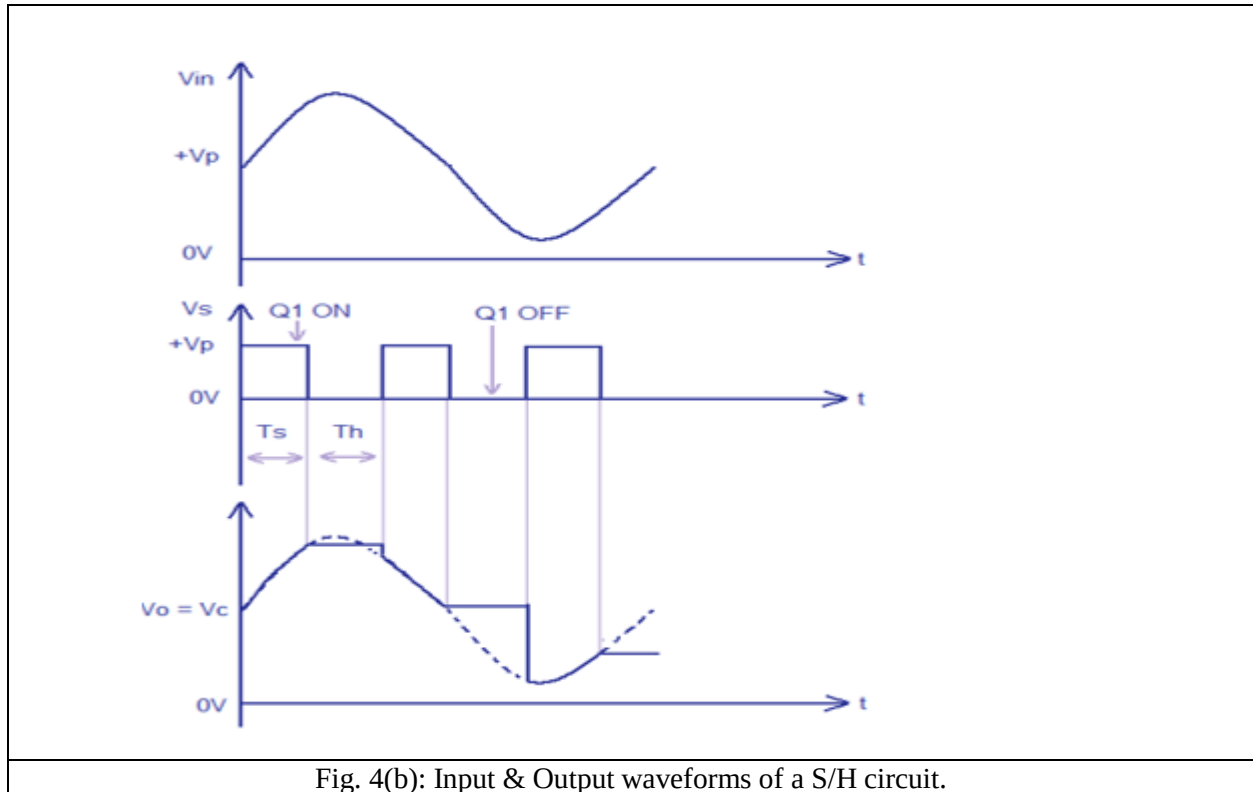


Fig. 4(b): Input & Output waveforms of a S/H circuit.

S/H circuits are used in the following applications:

- In connection with A/D converters (analog signal de-multiplexers), especially the successive approximation type, to hold the signal constant while the A/D conversion is taking place.
- As D/A converter output deglitches in CRT displays.
- Applications requiring fast acquisition-very low droop rate S/H circuit

EQUIPMENT:

1. DC Power Supply,
2. Function/Signal Generator,
3. Breadboard,
4. Resistor-1 k Ω (2 Nos), 100 k Ω (2 Nos),
5. Potentiometer-10 k Ω (2 Nos),
6. Diode (2 Nos),
7. Capacitor-0.01 μ F (2 Nos), 0.047 μ F (2 Nos),
8. 555 Timer IC,
9. Op-Amp (LM 741),
10. MOSFET IFR-540,
11. Oscilloscope, and
12. Connecting wires.

KNOWLEDGE PROFILE AND APPLICATION:

1. Basic Working Principle of 555 Timer,
2. Sample-and-Hold (S/H) Circuit, and
3. Effect of Sampling Frequency and Duty Cycle on the waveforms of the output of S/H circuit.

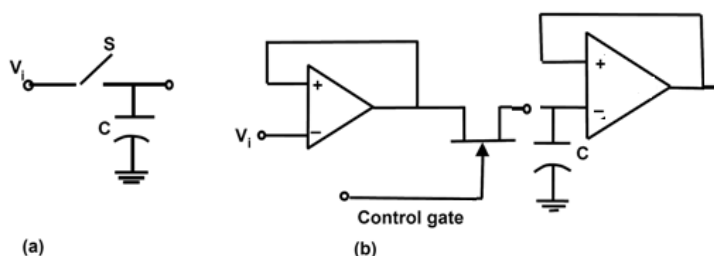
PROCEDURE:

1. Make the connection of 555 timers as shown in Fig. 4(c).
2. Observe the output waveshape at pin 3 of 555 timers and vary the duty cycle with the tuning of potentiometer. Choose a sampling frequency of 2 KHz.
3. Complete the connection for the whole circuit diagram of Fig. 4(c).
4. Apply a sinusoidal voltage of $5 V_{p-p}$ at the non-inverting input pin 3 of first Op-Amp of Fig. 4(c). Use a charging capacitor value of $0.1 \mu F$ in the circuit.
5. Observe the input sinusoidal signal on the oscilloscope at Channel 1 and output at pin 6 of second Op-Amp on Channel 2. Choose an operating frequency of 1 kHz, 500 Hz and 100 Hz for the input signal using the function generator and observe their effect on output signal. Comment on the difference.
6. Now set the operating frequency of the input signal at 100 Hz and vary the duty cycle of the 555 Timer IC output and observe their effect on the acquisition time and hold time of the output signal.
7. Now change the holding capacitor value to $0.033 \mu F$ and $0.047 \mu F$ and observe their effect on the output signal. Comment on your observation.
8. Now construct the circuit of Fig. 4(d). Use $100 k\Omega$ resistor in the feedback path. Repeat steps 1-7 above for the response of this circuit. Does the gain obtained from the circuit make the output response better in terms of acquisition time and hold time? Explain.

CIRCUIT DIAGRAM:

In this experiment, the response of two basic S/H circuit configuration will be studied and evaluated. These two circuits are given in Fig. 4(c) and Fig. 4(d). The 555 Timer IC used here produces the sampling pulse train to the gate of the MOSFET which acts as a switch. During the sample, MOSFET switch is turned on and the holding capacitor, C_H charges up to the level of the analog input voltage. At the end of this short sampling pulse, the MOSFET is switched off. This isolates the holding capacitor, C_H and the voltage across C_H and therefore V_o will essentially remain at the value of the input voltage sampled at the trailing edge of the sampling pulse. The acquisition time, t_{ac} of a S/H circuit is the time required for the holding capacitor to charge up to a level close to the input voltage during the sampling time. For the circuit configuration, there are three principal factors that will control the acquisition time.

- a) The RC time constant resulting from the resistance, $r_{dc(on)}$ of the MOSFET switch and the holding capacitor, C_H .
- b) Maximum output current which can be sourced and sunk by the first Op-Amp in the S/H circuit.
- c) The slew rate capability of the Op-Amp.



The output impedance of first Op-Amp is low, that's why there will be sufficient amount of current to flow across the MOSFET. The acquisition time, t_{ac} will be much faster and it will not be affected by the $r_{dc(on)}$ of the MOSFET. The second Op-Amp used in the circuit isolates C_H from the load to prevent rapid discharge. Due to the high input impedance of this second Op-Amp, the value charged by the holding capacitor, C_H cannot discharge and it remains constant during the MOSFET off period.

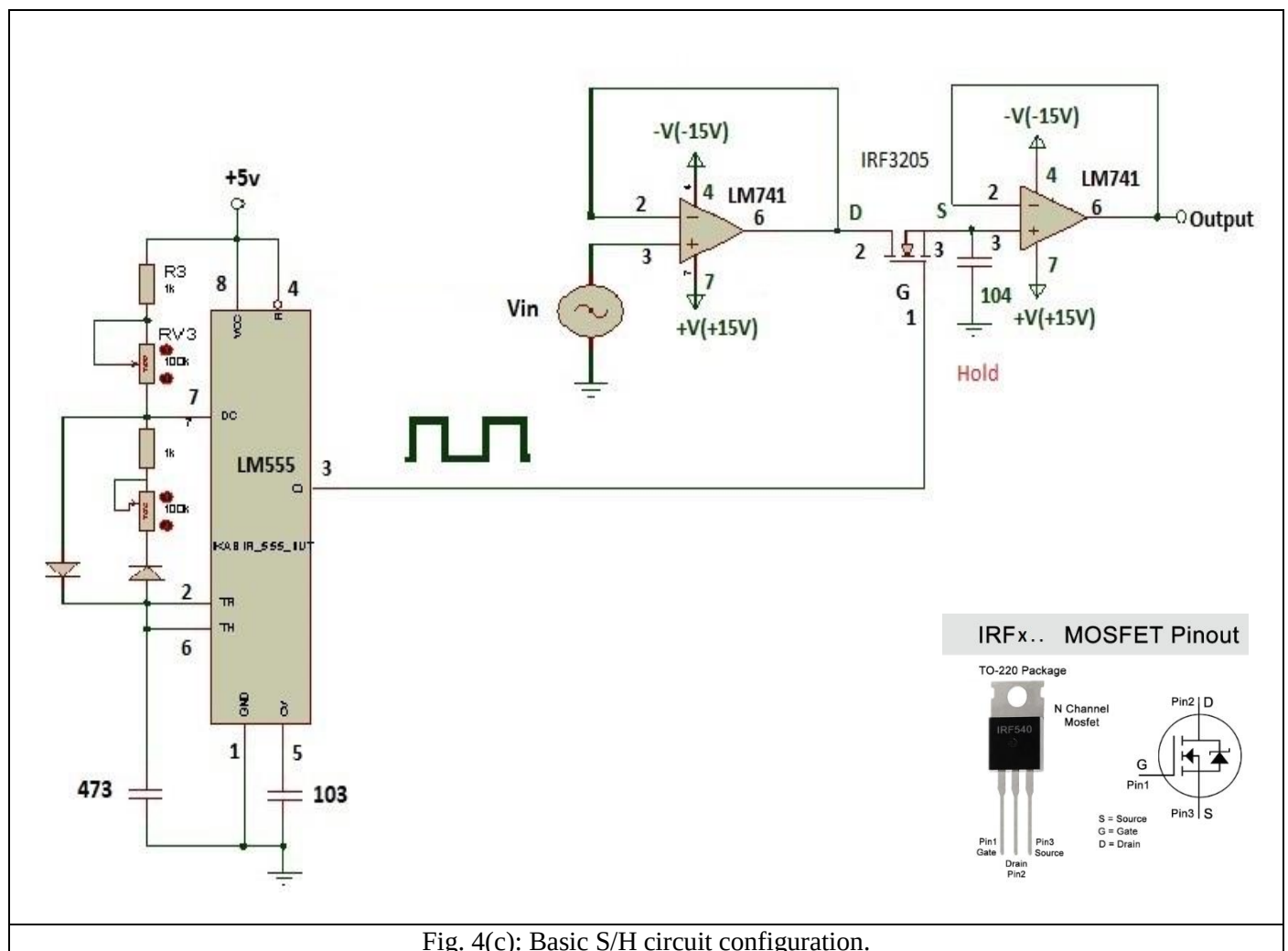
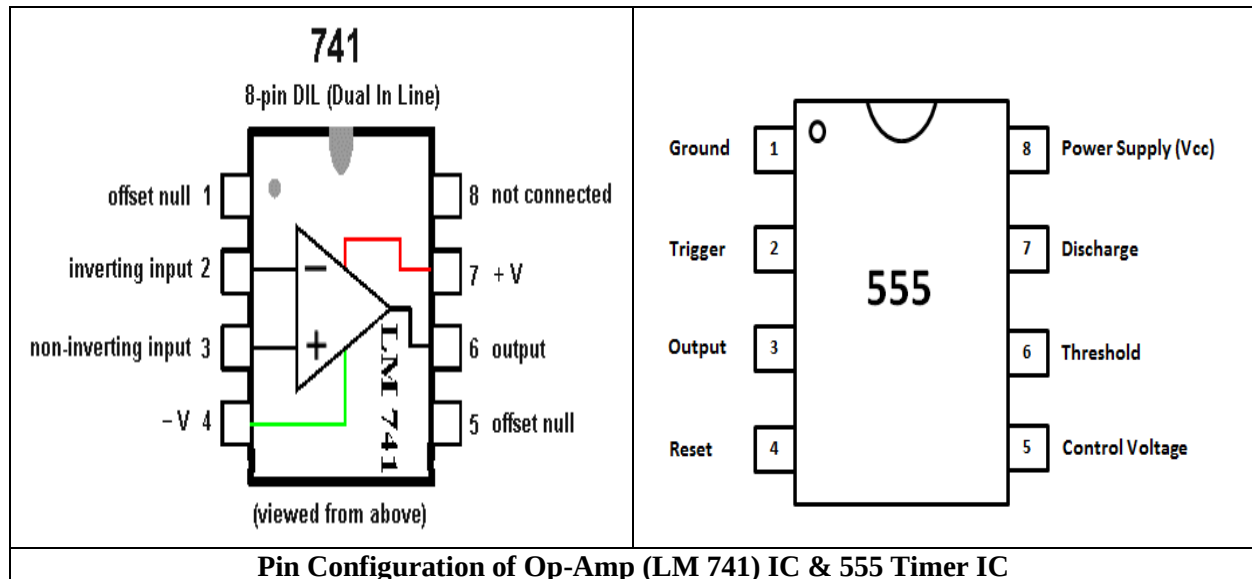


Fig. 4(c): Basic S/H circuit configuration.

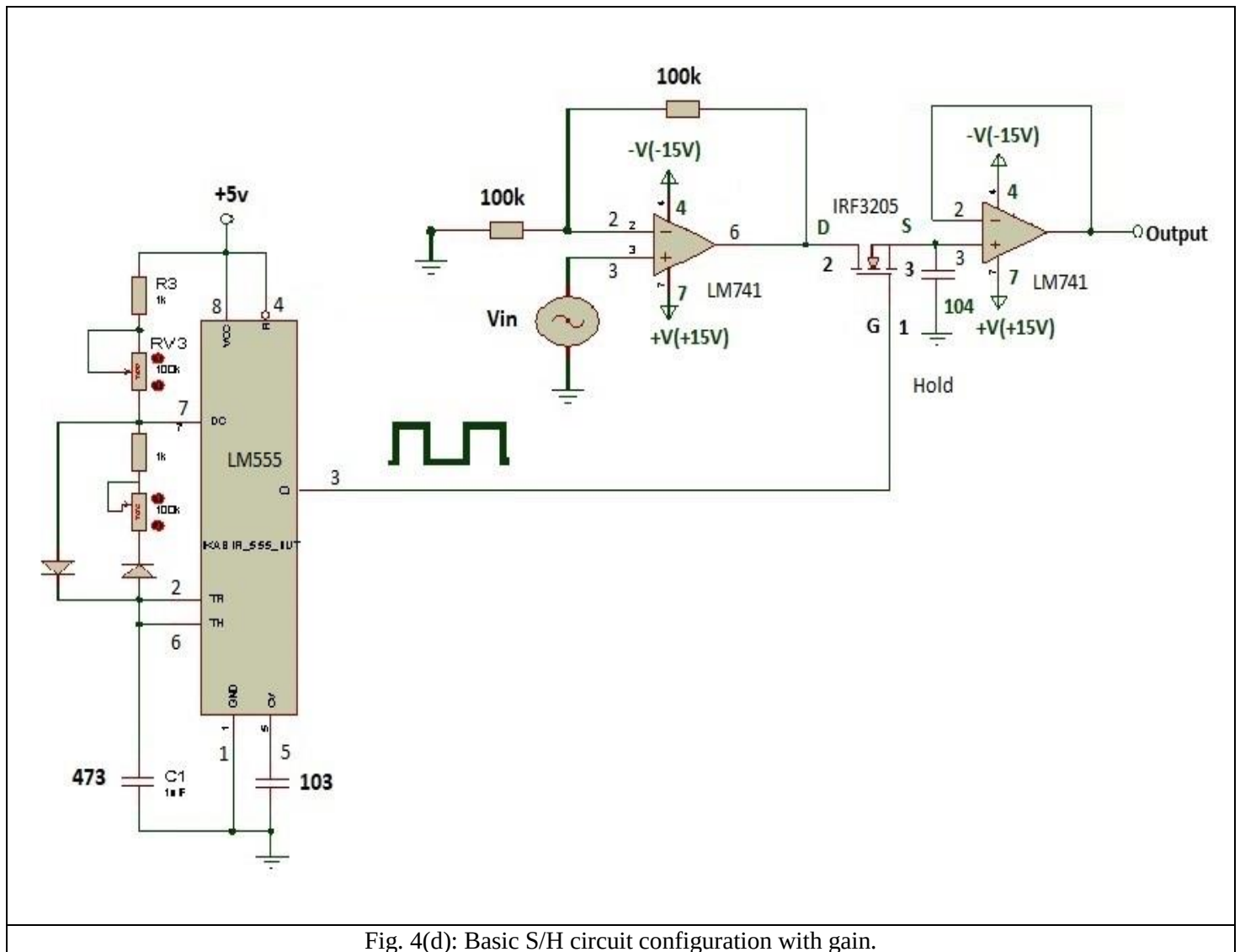


Fig. 4(d): Basic S/H circuit configuration with gain.

REPORT QUESTION:

1. Explain the differences between Fig. 4(c) and Fig. 4(d) along with the elaboration of working principle of Fig. 4(d).
2. Analyze the response of S/H circuit for different values of sampling frequency as well as different values of duty cycle.